

Refraction columns

All sphere, cylinder and MRSE fields use dioptres. Keep recorded and calculated MRSE separately. Derive MRSE only from valid paired sphere and cylinder at the same encounter. A trailing plus sign in refraction is a positive sign when the numeric context is clear; the same symbol in visual acuity is a letter modifier.

CSV column	Exact source header	Clean field and unit	Interpretation and rule
15	"pre-op_sphere diopters"	sphere_baseline_D	Signed sphere. PL or PI in this field means plano subject to documented convention, not perception of light.
16	"preop_cylinder Diopters"	cylinder_baseline_D	Signed cylinder; confirm cylinder convention.
17	"pre-op_mrse Diopters"	mrse_baseline_reported_D	Compare with sphere plus half cylinder. Missing reported MRSE may coexist with a valid derived value.
18	"postop_sphere after 1 month "	sphere_1m_label_D	Signed sphere at source one-month label.
19	"postop_cylinder after 1 month "	cylinder_1m_label_D	Signed cylinder at source one-month label.
20	"postop_mrse after 1 month "	mrse_1m_label_reported_D	Recorded MRSE; consistency check only.
21	"postop_sphere after 6 months "	sphere_6m_label_D	Signed sphere at source six-month label.
22	"postop_cylinder after 6 months"	cylinder_6m_label_D	Signed cylinder at source six-month label.
23	"postop_mrse after 6 months "	mrse_6m_label_reported_D	Recorded MRSE; consistency check only.
24	"postop_sphere after 12 months "	sphere_12m_label_D	Signed sphere at source twelve-month label; date unverified.
25	"postop_cylinder after 12 months"	cylinder_12m_label_D	Signed cylinder at source twelve-month label; date unverified.
26	"postop_mrse after 12 months "	mrse_12m_label_reported_D	Recorded MRSE; requires same-visit verification for endpoint analysis.